

# NullWire Phase 0 v2

## Control-Plane Trust and Multi-RPC Directory Validation

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### Abstract

We present an updated NullWire Phase 0 prototype for blockchain-coordinated mixnet messaging. The system keeps payloads off-chain, uses Solana as a decentralized control plane, and routes fixed-size messages through a three-hop sealed packet path with gateway mailbox delivery. The main update in this version is not a stronger anonymity claim. It is a stronger statement about control-plane trust. We identify routing table poisoning through compromised RPC views as a first-class risk and add a multi-RPC quorum validation path for client-side directory fetch before route construction. We also define route-admission hardening targets such as freshness-gated liveness checks and quorum-backed node admission. The result is still a prototype, but a less naive one.

## 1 Introduction

An anonymous transport system can fail before its cryptography is ever exercised. If the route is built from a poisoned control-plane view, the data plane starts from false premises.

That is the specific failure mode addressed in this revision.

NullWire Phase 0 still makes the same architectural choice:

- payloads stay off-chain;
- Solana acts as control plane only.

What changes is the trust model for directory fetch and route construction.

## 2 System Model

Phase 0 contains four runtime components:

| Component             | Responsibility                                                    |
|-----------------------|-------------------------------------------------------------------|
| Control-plane program | Registry state, stake state, reward bookkeeping                   |
| Relay nodes           | Decrypt one hop layer, apply delay, forward inner packet          |
| Gateway nodes         | Accept submissions, receive final delivery, store mailbox entries |
| CLI                   | Fetch directory, validate snapshots, build packets, send, poll    |

Each registry record currently includes:

- node identifier;

- node role;
- endpoint;
- transport public key;
- stake state;
- last-seen timestamp; and
- reputation placeholder.

### 3 Threat: RPC-Poisoned Route Construction

The updated threat statement is:

If the client trusts one RPC endpoint, the control plane can be poisoned upstream of packet construction.

Examples include:

- substituting attacker-controlled relay endpoints;
- hiding honest nodes from the route candidate set;
- rewriting transport keys; and
- surfacing stale or quarantined nodes as valid route options.

This does not directly defeat the layered packet format. It defeats the assumptions used to build the route in the first place.

### 4 Current Mitigation

The Phase 0 CLI now supports:

- fetching the registry from multiple independent RPC endpoints;
- normalizing the resulting directory snapshots;
- requiring k-of-n agreement before accepting the view; and
- optionally filtering route candidates by `last_seen_unix` freshness.

This is a client-side mitigation, not an on-chain proof system.

## 5 Packet Construction

The data plane remains a sealed multi-hop packet path:

- fixed-size payload body;
- layered encrypted hop directives;
- compact hop labels rather than raw endpoint URLs in the Sphinx address field; and
- mailbox-only disclosure at the final gateway.

The current implementation uses a real Sphinx packet path for relay forwarding.

## 6 What Is Still Missing

Multi-RPC quorum validation is necessary, but not enough.

The next missing pieces are:

- recent uptime proofs that are stronger than a plain timestamp;
- validator-backed admission before a new node becomes route-eligible;
- auditable quarantine or suspension rules for nodes with bad behavior; and
- route selection policies that do more than pick the first relay per layer.

## 7 What We Borrow and What We Do Not

Validator-enforced security networks such as SafeNet are useful here only as a design lesson. The relevant lesson is:

- do not trust one view of intent or one view of state;
- move critical checks into a verifiable path; and
- make validator activity publicly auditable.

The wrong lesson would be to copy a transaction firewall directly into a mixnet data plane. NullWire is not defending execution semantics. It is defending route integrity and privacy-preserving transport.

## 8 Explicit Limitations

This version still does not prove:

- meaningful anonymity set size;
- resistance to global passive adversaries;
- production-safe node admission;
- censorship resistance under cartelized validators; or
- final packet cryptography beyond the current prototype scope.

## 9 Conclusion

The strongest honest statement about this revision is:

NullWire Phase 0 is a blockchain-coordinated mixnet messaging prototype that now treats control-plane trust as part of the security problem, not as an invisible assumption.

### Source Note

This revision was motivated in part by the broader security lesson described in:

<https://www.bankless.com/read/the-1-4b-hack-that-safenet-was-built-to-prevent>

The specific multi-RPC quorum and route-admission design described here is a NullWire adaptation, not a direct feature of that system.